Resume

Minwoo Lee

Bachelor Student, Kangnam University — Yongin, South Korea
Email: minwool0357@gmail.com | Phone: +82-10-6757-3689 | GitHub: github.com/minuum
Portfolio: minuum-portfolio | Research Page: minuum.github.io/MoNaVLA

EDUCATION

Kangnam University — Yongin, South Korea
Bachelor of Science in Artificial Intelligence (Senior, 1st Semester)
- Expected Graduation: Feb. 2027
- GPA: **4.27 / 4.5** (Ranked top of the department)

RESEARCH INTERESTS

- Vision-Language-Action (VLA) Models for Robotics
- Embodied AI & Mobile Robot Navigation
- Sim-to-Real Transfer & Physical Latency Mitigation
- Deep Reinforcement Learning

PUBLICATIONS & MANUSCRIPTS

- **EdgeGround-VLA: On-Device Goal-Directed Navigation via Dual Frozen Grounding Encoders**
  Journal of The Korea Society of Computer and Information (JKSCI), KCI Registered, **Under Review (2026)**.
  Minwoo Lee (1st Author / Project Lead), et al.
    - Fuses dual frozen grounding encoders (OWLv2 + Kosmos-2) with a 0.866M-parameter action head (1.128M total trained parameters).
    - Achieves **95.0% real-robot goal-reaching success rate** fully on-device on Jetson Orin NX.
    - Project Page: minuum.github.io/EdgeGroundVLA
- **Designing Conversational AI Services Using RAG-Based Loneliness Analysis for Koreans**
  Journal of the Korea Convergence Society (JCCT), KCI Registered, 2024.
  Minwoo Lee (2nd Author), et al.

CONFERENCES

- Presented at the International Conference on IPACT 2024 —
  Outstanding Paper Award
  Minwoo Lee (2nd Author), et al.

RESEARCH EXPERIENCES

Undergraduate Research Assistant — Advised by Prof. Inyeop Choe, Kangnam University
Project: MoNaVLA (Mobile Navigation Vision-Language-Action Models) | *Sep. 2025 – Present*
- **Diagnosis & Analysis:** Diagnosed structural text-attention collapse in the Kosmos-2 VLA backbone through per-layer attention measurement and frozen linear probe classification (val_acc: 96.6%), revealing that post-training had silently destroyed the text pathway regardless of downstream fine-tuning. - **Insight Validation:** Confirmed that visual grounding drives navigation decisions via a target object masking ablation that induced 100% action reversal. - **Architecture Design:** Designed a two-stage decomposition pipeline integrating PaliGemma2 as a zero-shot visual grounder with the Kosmos-2 vision encoder, bounding box history, and L2 normalization. - **Performance:** Improved closed-loop navigation success rate from 10.3% (simple MLP baseline) to 66.7% (Decomposition v1) to **96.6%** (final, FPE: 0.094m) — a 9.4x gain from pipeline refinement alone on an identical grounding source, recovering from near-zero success in the end-to-end Kosmos-2 baseline. - **Real-Robot Validation:** Deployed the PaliGemma2 grounder on the physical Serbot 2 platform, raising real-robot grounding accuracy from 0% to 51.4% across 6 sessions (512 frames) by fixing camera-decoding bugs and applying bounding-box noise filtering; full closed-loop control testing is the next step. - **Award:** Won the **Silver Prize at the 2026-1 Kangnam University Capstone Design Competition**.

AWARDS AND HONORS

- **Silver Prize**, 2026-1 Kangnam University Capstone Design Competition
- **Grand Prize (Dean of College of Engineering Award)**, 2025 Kangnam University Hackathon 'Kangnengthon' (Jointly hosted by Kangnam University & GDGoC: Kangnam University)
- **Outstanding Paper Award**, IPACT 2024 International Conference
- **First Prize (Best Project Award)**, Software Academic Festival (Kangnam University, 2024)
- **Excellence Award**, AI Academic Festival (Kangnam University, 2024)
- **Award**, K-HTML Hackathon (2024)

PROJECTS

Serbot 2 Robot Platform Integration — Individual Project | *Jan. 2025 – Mar. 2025*
- Configured a Jetson Orin-based Linux environment and integrated ROS2 driver nodes for the Serbot 2 omnidirectional mobile robot platform. - Implemented core locomotion control scripts and configured the hardware-software communication interface to host VLA inference models. - Resolved physical-to-virtual command delays and stabilized actuator control signals.

Ladi — Software Academic Festival | *Oct. 2024*
- Developed an AI-based healthcare routine recommendation system (**Ladi**), winning the **First Prize**. - Implemented a hybrid search system combining dense (FAISS) and sparse (KiwiBM25) retrievers to optimize medical QA search. - Integrated a CrossEncoder Reranker to improve recommendation quality, and optimized prompt templates using LangSmith. - Preprocessed AIHub healthcare QA data to categorize diseases and route intents.

Liberty — AI Academic Festival | *May 2024*
- Served as Project Manager (PM) and lead architect for an AI Legal QA Agent system (**Liberty**), winning the **Excellence Award**. - Built a multi-agent workflow using LangGraph, integrating dynamic query rewriting and quick filtering. - Implemented a hybrid retriever (FAISS + KiwiBM25 + Pinecone) achieving 1.000 score in faithfulness and retrieval accuracy benchmarks.

SKILLS AND TECHNIQUES

- **AI/ML:** PyTorch, HuggingFace, LoRA/PEFT, PaliGemma2, CLIP, SigLIP, Kosmos-2, LangGraph (Agent), LangSmith, FAISS, Pinecone (Vector DB), BM25, CrossEncoder (Reranker)
- **Robotics & Infra:** ROS2, Jetson Orin, Linux (Ubuntu), Locomotion Control, Asynchronous Control, Git
- **Programming Languages:** Python, C++, SQL
- **Languages:** Korean (Native), English (Intermediate — preparing for TOEIC)

COURSEWORK

- **Deep Reinforcement Learning** — *Self-study (UC Berkeley CS285: Deep RL course lectures & homework)*
  - Focused on Imitation Learning, Policy Gradients, Actor-Critic methods, and value-based RL.